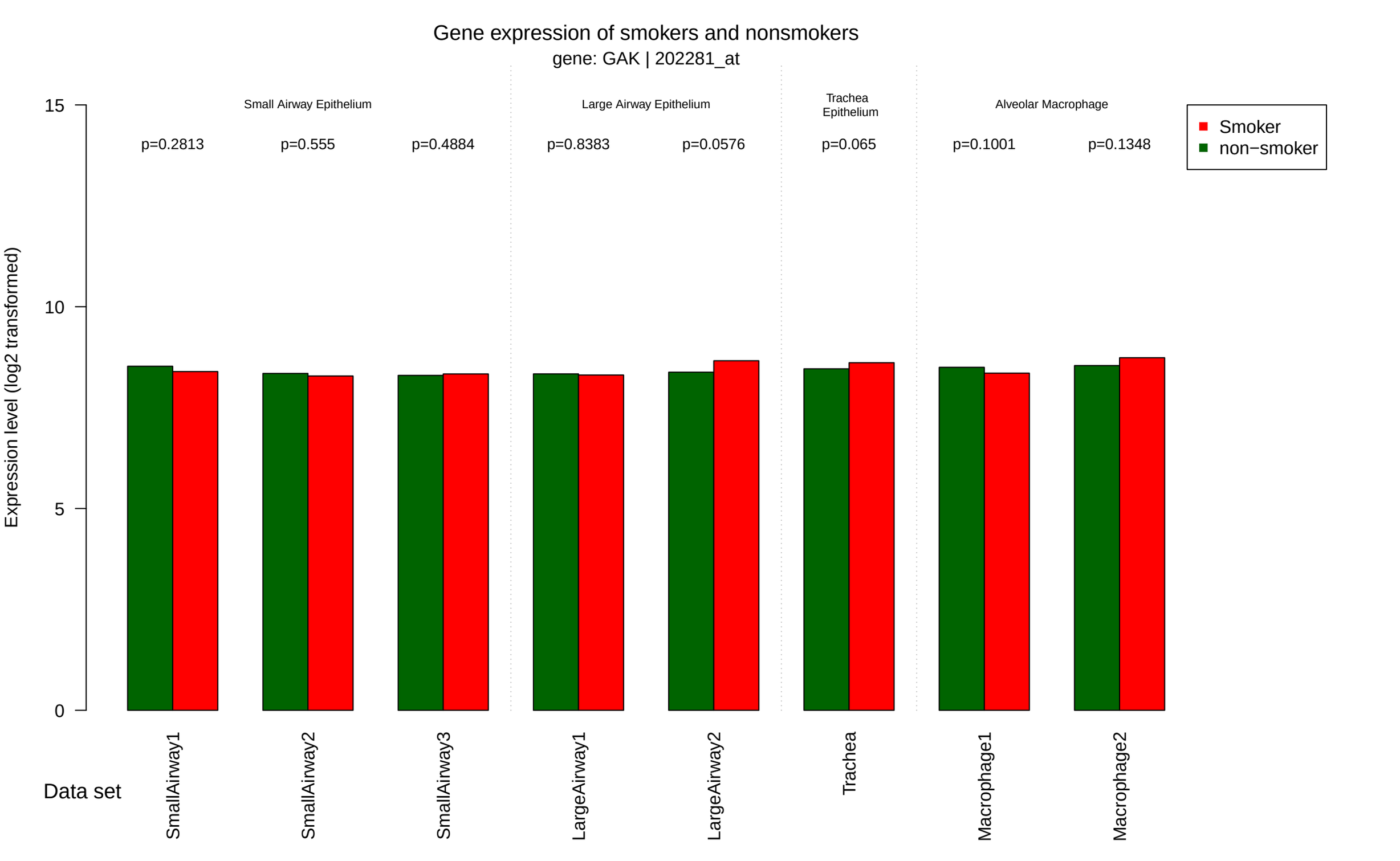
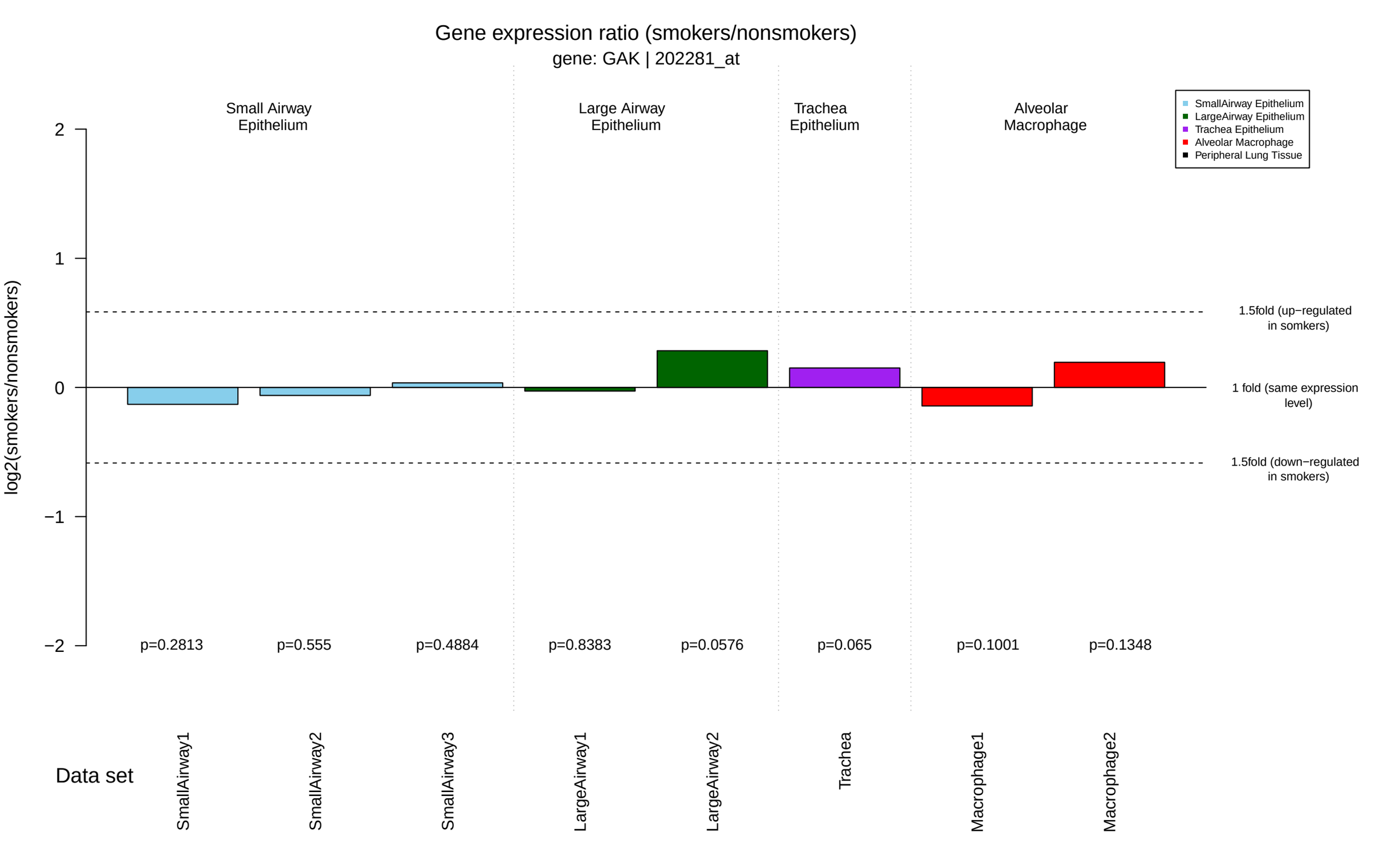
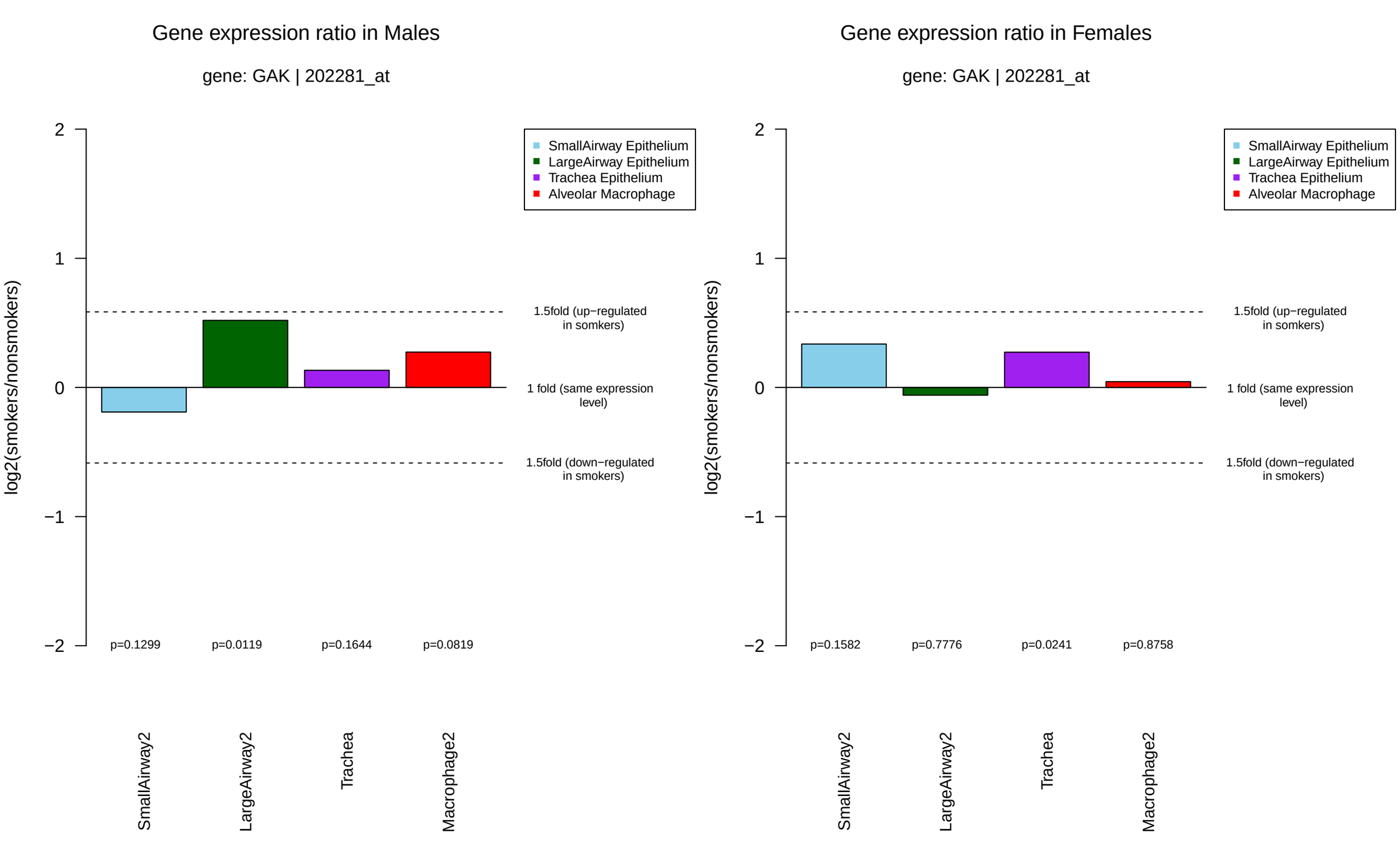
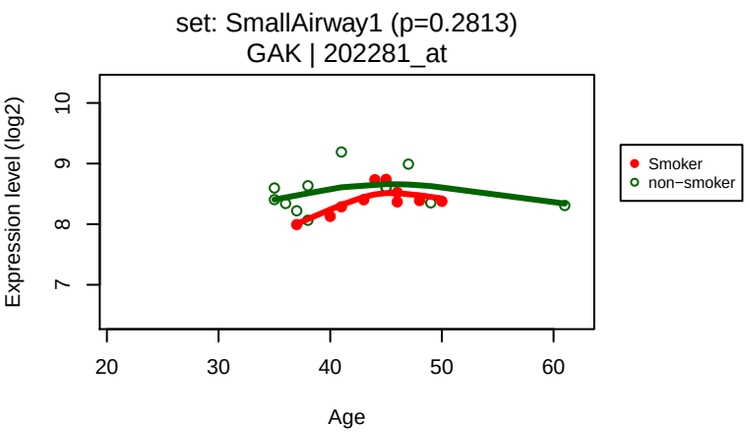
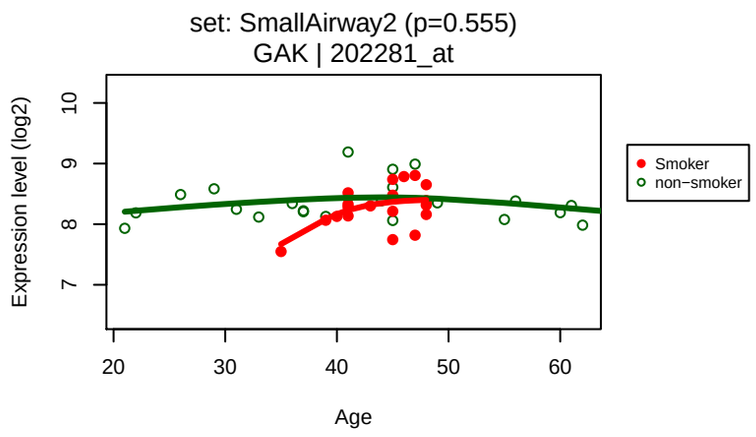
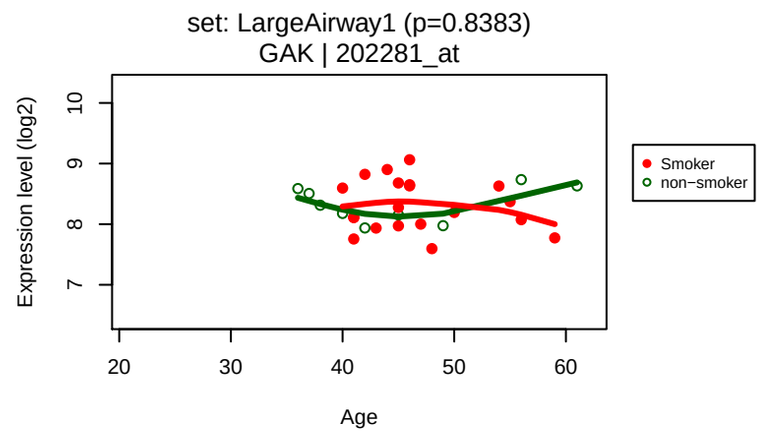

Gene expression of smokers and nonsmokers
gene: GAK | 202281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2813
p=0.555
p=0.4884
p=0.8383
p=0.0576
p=0.065
p=0.1001
p=0.1348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

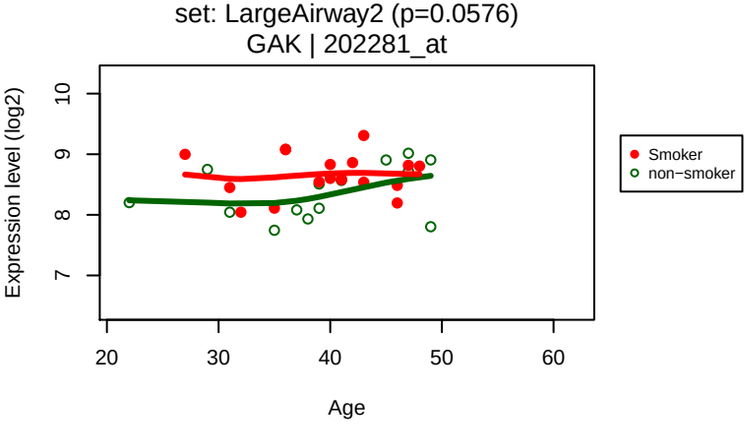

Gene expression ratio (smokers/nonsmokers)
gene: GAK | 202281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2813
p=0.555
p=0.4884
p=0.8383
p=0.0576
p=0.065
p=0.1001
p=0.1348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

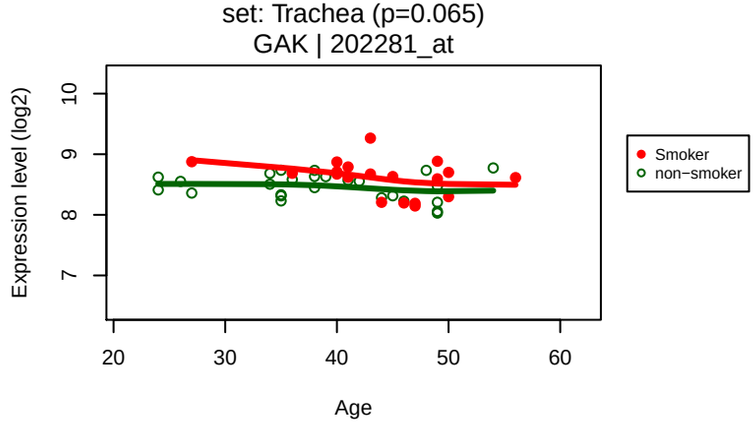

Gene expression ratio in Males
gene: GAK | 202281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1299
p=0.0119
p=0.1644
p=0.0819
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAK | 202281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1582
p=0.7776
p=0.0241
p=0.8758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

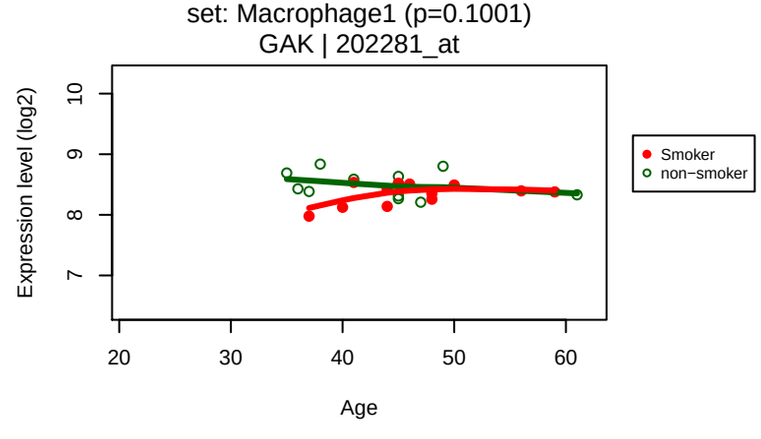

set: SmallAirway1 (p=0.2813)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker

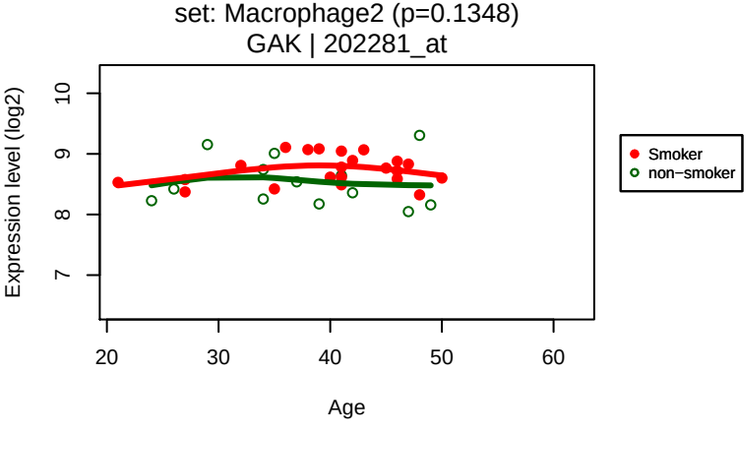

set: SmallAirway2 (p=0.555)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker

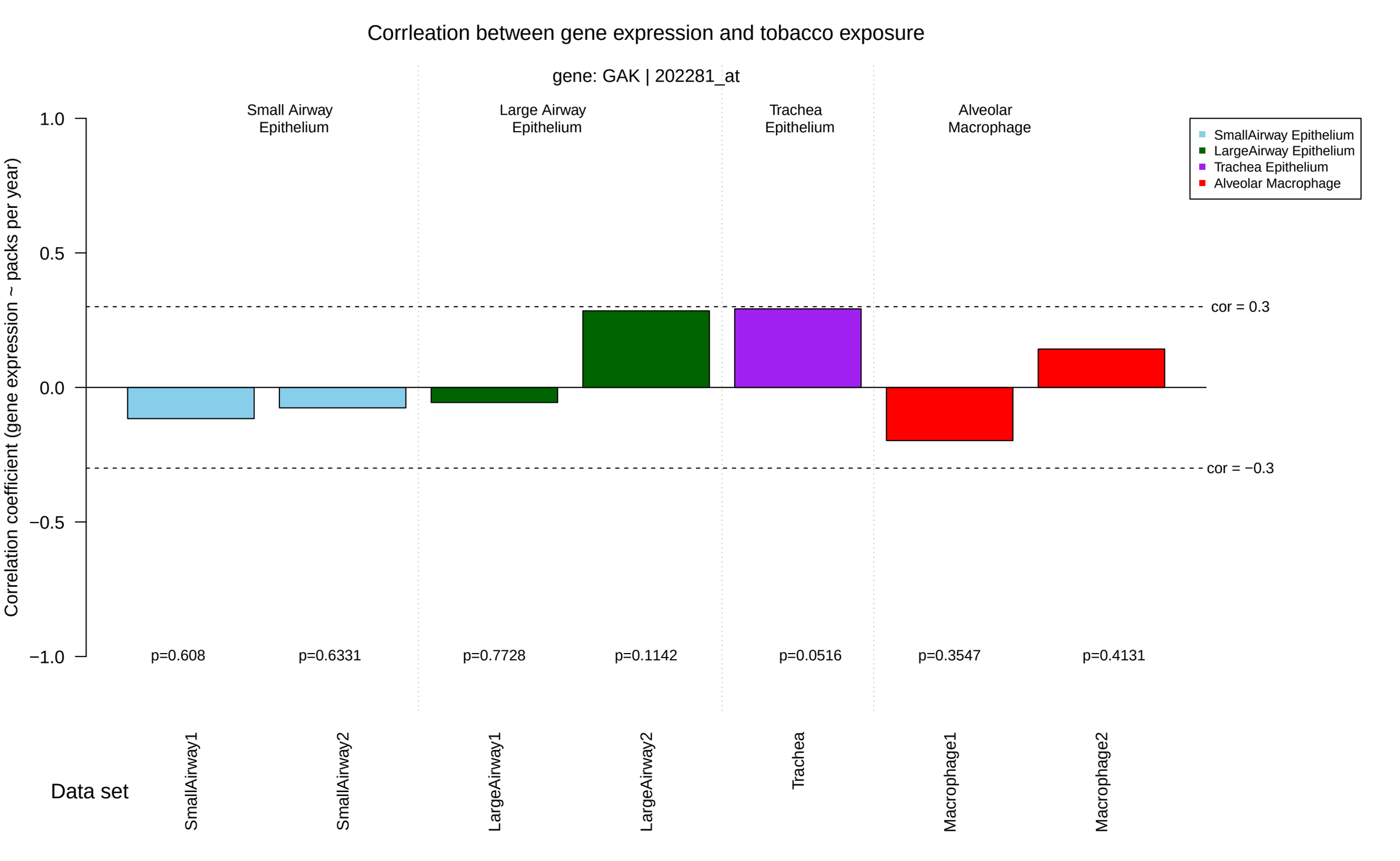

set: LargeAirway1 (p=0.8383)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0576)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.065)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1001)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1348)
GAK | 202281_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAK | 202281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.608
p=0.6331
p=0.7728
p=0.1142
p=0.0516
p=0.3547
p=0.4131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2